DAY 05

ETHERNET & MAC ADDRESSING

Networking Fundamentals Assignment

Student Name	Nethra M
Assignment Code	DAY05-ETHERNET-MAC
Total Marks	20 Marks

OBJECTIVE

To understand Ethernet communication, Ethernet frames, MAC addressing, switch learning and forwarding, and identify local network configuration using Windows command-line tools.

PART A: THEORY (10 MARKS)

1. What is Ethernet?

Ethernet is a widely used networking technology that connects devices within a Local Area Network (LAN) and enables communication over wired network connections.

2. What is an Ethernet frame?

An Ethernet frame is a data unit transmitted over Ethernet. It contains destination and source MAC addresses, a type/length field, payload, and a Frame Check Sequence (FCS) used for error detection.

3. What is a MAC address?

A MAC address is a link-layer address associated with a network interface and used for local frame delivery. A common Ethernet MAC address is 48 bits long and written in hexadecimal, for example, `00-1A-2B-3C-4D-5E`.

4. What does MAC stand for?

MAC stands for **Media Access Control**.

5. Difference between a MAC address and an IP address?

MAC address: Used for local Ethernet frame delivery at OSI Layer 2.
IP address: A logical network-layer address used to route packets between networks at OSI Layer 3. MAC addresses are usually hexadecimal; IPv4 uses dotted-decimal notation and IPv6 uses hexadecimal groups.

6. What is a MAC Address Table?

A switch table that records learned MAC addresses and the associated switch ports, usually within a VLAN, so frames can be forwarded efficiently.

7. How does a switch learn MAC addresses?

A switch reads the source MAC address of an incoming frame and associates it with the port where the frame arrived and the relevant VLAN. Entries may age out if not refreshed.

8. What is a source MAC address?

The source MAC address identifies the interface that sent the Ethernet frame on the local network segment.

9. What is a destination MAC address?

The destination MAC address identifies the intended recipient of an Ethernet frame. It can identify an individual interface or a multicast/broadcast destination.

10. What is an unknown unicast frame?

A frame whose destination MAC address is not in the switch forwarding table for the relevant VLAN. The switch typically floods it through eligible ports in that VLAN, excluding the incoming port.

11. What is flooding?

Flooding forwards a frame out of multiple eligible switch ports in the same VLAN, excluding the port where it arrived. It is commonly used for broadcast and unknown-unicast frames.

12. What is known unicast forwarding?

When the destination MAC address is in the switch table, the switch forwards the frame only through the associated port instead of flooding it.

13. At which OSI layer does Ethernet switching primarily operate?

Ethernet switching primarily operates at **Layer 2 — the Data Link Layer**.

14. Why does a switch maintain a MAC Address Table?

It maps learned MAC addresses to switch ports, allowing efficient forwarding and reducing unnecessary traffic on other ports.

15. Why is Ethernet knowledge important in cybersecurity?

It helps security professionals understand local network traffic, identify interfaces, analyze packet captures, troubleshoot connectivity, and investigate suspicious activity.

PART B: PRACTICAL IDENTIFICATION (4 MARKS)

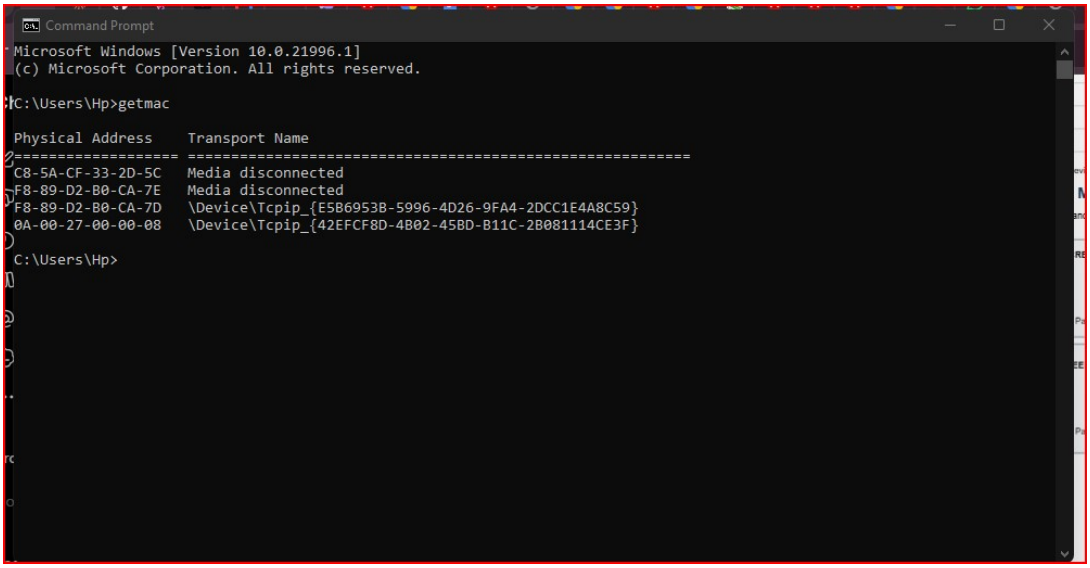
The following values were read from the screenshots you provided. The Ethernet adapter shown as media disconnected is not the active adapter in that screenshot.

Required detail	Observed result
Adapter name	VirtualBox Host-Only Ethernet Adapter (Ethernet 2)
MAC address / Physical Address	0A-00-27-00-00-08
IPv4 address	192.168.56.1
Default gateway	Not shown / blank for this adapter
Other adapter	Realtek PCIe GbE Family Controller — Media disconnected

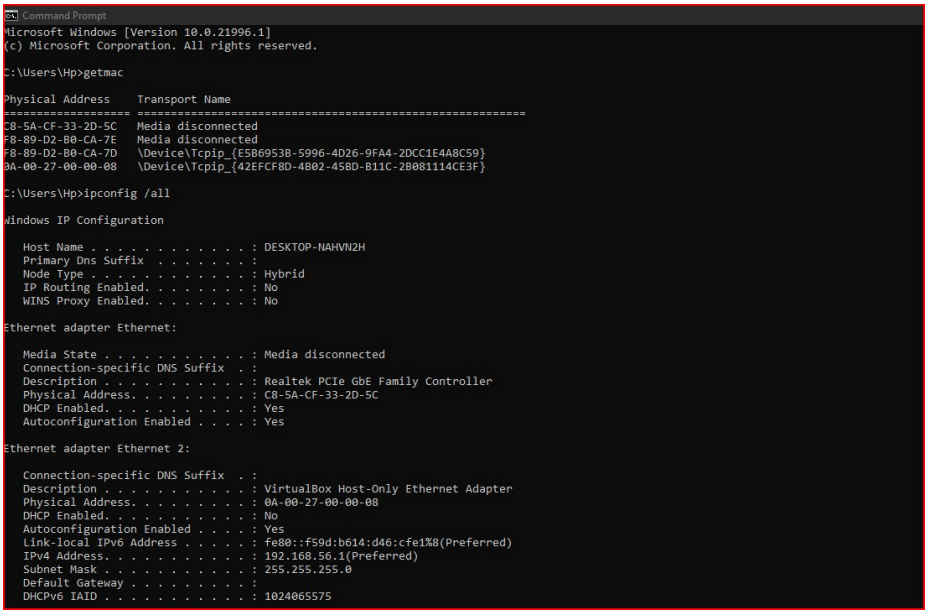
Note: The screenshots show a VirtualBox host-only adapter with 192.168.56.1 and no default gateway. The ARP output also shows an interface at 10.72.32.74, but the supplied ipconfig screenshot is cut off before its full adapter details. If your instructor expects the active internet-connected adapter, run `ipconfig /all` again and capture the complete Wi-Fi/Ethernet section.

PART C: SCREENSHOTS (3 MARKS)

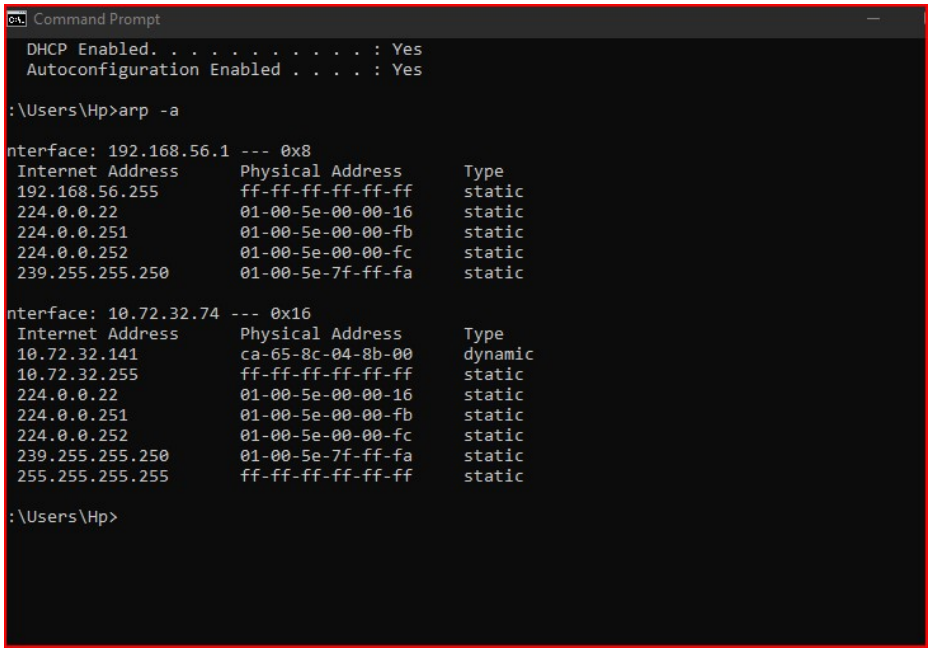
Screenshot 1 — getmac



Screenshot 2 — ipconfig /all



Screenshot 3 — arp -a



PART D: REPORT QUALITY (3 MARKS)

Submission checklist

- ■ Name (Nethra M) and Day 05 are shown on page 1.
- ■ All 15 theory questions are answered.
- ■ Practical details are recorded from the supplied command output.
- ■ Screenshots of getmac, ipconfig /all, and arp -a are attached and readable.
- ■ Formatting and spelling have been checked.

CONCLUSION

This assignment strengthened my understanding of Ethernet frames, MAC addresses, and Layer 2 switching. I learned how switches build MAC address tables from incoming source addresses and forward known unicast frames or flood frames when the destination is unknown. The commands getmac, ipconfig /all, and arp -a helped me inspect adapter information and local address mappings.

Safety: Commands were run for learning on the provided Windows device/network environment.